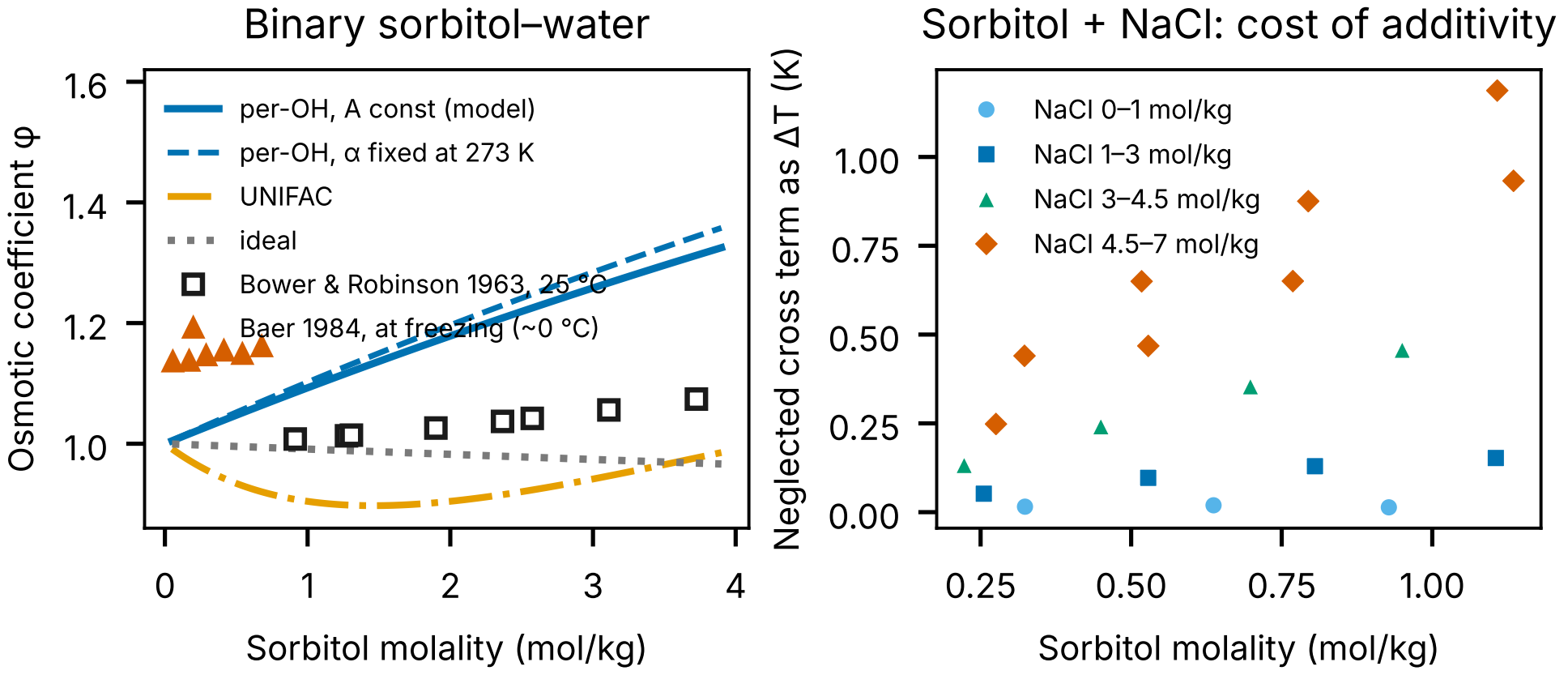
Binary sorbitol–water
Osmotic coefficient φ
1.6
1.4
1.2
1.0
per-OH, A const (model)
per-OH, α fixed at 273 K
UNIFAC
ideal
Bower & Robinson 1963, 25 °C
Baer 1984, at freezing (~0 °C)
0
1
2
3
4
Sorbitol molality (mol/kg)
Sorbitol + NaCl: cost of additivity
Neglected cross term as ΔT (K)
1.00
0.75
0.50
0.25
0.00
NaCl 0–1 mol/kg
NaCl 1–3 mol/kg
NaCl 3–4.5 mol/kg
NaCl 4.5–7 mol/kg
0.25
0.50
0.75
1.00
Sorbitol molality (mol/kg)